MILITARY SPECIFICATION

TUBING, STERILIZATION

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers heat sealable plastic tubing for packaging medical supplies for gas sterilization and storage.

1.2 Classification.

1.2.1 Sizes. Tubing shall be of the following lay-flat widths as specified (see 6.2):

Size I.	3 inches.
Size II.	5 inches.
Size III.	10 inches.
Size IV.	20 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Varporproof or Waterproof, Flexible.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D257	D-C Resistance or Conductance of Insulating Materials.
ASTM D374	Thickness of Solid Electrical Insulation.
ASTM D882	Tensile Properties of Thin Plastic Sheeting.
ASTM D1434	Gas Transmission Rate of Plastic Film and Sheeting.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The tubing shall be extruded polyethylene film modified with a low percentage of ethyl vinyl acetate to afford a broader temperature range for heat sealing, and tinted green to identify it with gas sterilization. The film shall be treated with an internal antistatic additive such that 1 x 10^{13} ohms resistance, maximum, shall be measured when tested as specified in 4.4.1. The minimum values for tensile strength and elongation listed in Table I shall apply when tested as specified in 4.4.2.

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Table I.

Physical Property	Machine Direction	Cross Direction
Tensile breaking strength (pounds per inch of width)	4	3.5
Ultimate elongation (percent)	100	500

A single thickness of film shall be transparent enough to easily read pica type through it. A heavy duty resin shall be used in the low density (0.918) copolymer for the tubing to give it high impact and tear strength, and high slip anti-block characteristics to permit easy loading.

3.1.1 Permeability. The finished tubing material shall be such that ethylene oxide gas permeability at 70°F and 29.92 inches of Hg is a minimum of 10,000 cubic centimeters, per mil thickness, per square meter, per 24 hours-when tested as specified in 4.4.3.

3.2 Design. The sterilization tubing shall be extruded, lay flat, polyethylene, tubular film, firmly wound on a paperboard reel. Tubing shall be the width specified in 1.2.1 for the indicated size. The tubing dimensions and tests indicated in Table II shall apply.

Table II.

Dimension	Tolerance	Test paragraph
Widths		
3 inches and 5 inches	± 1/8 inch	4.4.4
10 inches and 20 inches	± 1/4 inch	4.4.4
Length each roll, 75 feet	70 feet, min.	4.4.5
Thickness, single film, 0.0030 inch	± 0.0005 inch	4.4.6

More than one tubing break shall be cause for rejection (see 4.4.5).

3.2.1 Reel. The reel on which the tubing is wound shall be a paperboard tube with an internal diameter of 3 inches, ± 1/8 inch, and a minimum wall thickness of 1/8 inch.

3.2.2 Tubing roll container. Each roll shall be contained in a chipboard or corrugated paperboard box with a single flap top. Construction shall be such that the film may be unrolled directly from the box, without opening it, when the film is passed between the single flap top and the adjacent side wall.

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3.2.3 Instructions for use. Instructions for using the tubing shall be provided with each roll including directions that tubing should not be heat sealed. Included with the instructions shall be the following information on chemical resistance.

* 3.2.3.1 Chemical resistance. The tubing will not be affected by prolonged contact with oils, fats, waxes, greases, weak acids, weak or strong alkalies, and organic solvents at room temperature, except for the antistatic or electrical properties which may change. Polyethylene will be attacked by strong acids and chlorinated and aromatic solvents at temperatures over 122°F. However, tubing which has been in contact with any of these chemicals should not be used to package medical items for gas sterilization. The contaminating chemical might transfer to the item being sterilized making that item unfit for use; or, toxic by-products might be produced by a chemical reaction between ethylene oxide and the contaminating chemical.

3.3 Workmanship. Workmanship shall be first class throughout. The tubing shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the tubing may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and Table III.

Table III.

	Inspection level	AQL (defects per hundred units)
For visual examination Major defects	II	1.0
Total defects (Major and Minor combined)	II	4.0

4.2.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105 and Table IV. The sample for examination purposes shall be one roll.

Table IV

Characteristic	Requirement	Test Procedure	Number of Samples per Lot*
Antistatic resistance	3.1	4.4.1	3
Physical properties:			
Tensile strength	3.1	4.4.2	2
Ultimate elongation	3.1	4.4.2	2
Permeability	3.1.1	4.4.3	1
Dimensions:			
Width	3.2	4.4.4	5
Length	3.2	4.4.5	3
Thickness	3.2	4.4.6	5

*The same sample may be used for more than one test. Acceptance level shall be zero for all sample sizes.

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4.3 Examination. The tubing shall be examined to determine compliance with this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects in Table V.

Table V.

Categories	Defects*
Major	
101	Tubing not free of pinholes, tears, cracks.
102	Tubing not free of wrinkles, bunching.
103	Tubing not free of streaks.
104	Tubing not firmly and evenly rolled on the reel.
Minor	
201	Finished roll not round.
202	Paperboard reel extends beyond roll on either end.
203	Tubing not free of dirt, grease, loose foreign matter.
204	Tubing does not unroll freely from box.
205	Instructions for use not provided.

*Examination shall not be restricted to the classified defects listed above.

4.4 Tests. Tests shall be conducted to determine compliance with this specification.

4.4.1 Antistat resistance. Three 6-inch tubing specimens cut from near the beginning, middle, and end of the roll shall be tested in accordance with ASTM D257. The average resistance of the three specimens shall be recorded.

4.4.2 Physical properties. Tensile strength and ultimate elongation shall be measured in accordance with ASTM D882, Method A, using specimens from two sample rolls.

4.4.3 Permeability. Three specimens per sample roll, taken from anywhere in the roll after discarding the first three layers, shall be tested in accordance with Method M or Method V of ASTM D1434.

4.4.4 Tubing width. At any point in the roll, other than a cut edge, the tubing's lay flat width shall be measured to the nearest 1/16 inch.

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4.4.5 Tubing length. The length of the tubing shall be measured to the closest inch. If either end of the roll is cut irregularly or on an angle, the measurement shall not include that portion for which there is not a full width of tubing perpendicular to the long axis of the tubing. Also, the number of tubing breaks, if any, shall be noted.

4.4.6 Film thickness. Single film thickness to the closest 0.0001 inch shall be recorded in accordance with ASTM D374, Method A or Method C. Discarding the three outer wraps of the roll, three readings shall be taken on each specimen, measured over a minimum length of 12 inches at three equally spaced points, and averaged.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One roll, of the size specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table VI shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table VI.

Size	Unit package	Intermediate package	Exterior container
I	1 unit	4 units	24 units
II	1 unit	4 units	16 units
III	1 unit	4 units	16 units
IV	1 unit	Not required	4 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged into a dispensing type box of appropriate size and design. Closure shall be adequate to prevent accidental opening.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic, except commercial colors will be acceptable. Closure shall be adequate to prevent accidental opening.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with paragraph 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method I of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The sterilization tubing shall be for use by the military medical services, in various widths and cut to the length required, as a heat sealable package for gas sterilization and storage of medical and dental supplies.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).
(c) Size of tubing required.

6.3 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
	TUBING, STERILIZATION, Polyethylene, 0.003 inch thick, 75 feet.
6530-00-104-7552	3 inches Wide Flat.
6530-00-104-7553	5 inches Wide Flat.
6530-00-104-7564	10 inches Wide Flat.
6530-00-104-7584	20 inches Wide Flat.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DSA-DM

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Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

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3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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